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United States Patent	12409330
Kind Code	B2
Date of Patent	September 09, 2025
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Airtight device and feedthrough module

Abstract

An airtight device includes a tank, a sink and a feedthrough module. The sink is disposed inside the tank, and an opening is formed on the sink. The feedthrough module is disposed on the opening. The feedthrough module includes a base, a sealing component, a covering component, a transmission component and a plurality of fixing components. A groove is formed on the base. A part of the sealing component is disposed inside the groove. The covering component is assembled with the base and adapted to press the sealing component. A plurality of fixing holes is formed on the covering component. The transmission component is assembled with the covering component. The plurality of fixing components is adapted to insert into the plurality of fixing holes and engage with the base for pressing the sealing component by shortening a distance between the base and the covering component.

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Appl. No.: 18/098714

Filed: January 19, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230158317 A1	May. 25, 2023

Foreign Application Priority Data

TW	109129348	Aug. 27, 2020
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Related U.S. Application Data

continuation-in-part parent-doc US 17109071 20201201 US 11905747 child-doc US 18098714

Publication Classification

Int. Cl.: A61N1/375 (20060101); G02B6/42 (20060101)

U.S. Cl.:

CPC A61N1/3754 (20130101); G02B6/4248 (20130101);

Field of Classification Search

CPC: A61N (1/3754); G02B (6/4248)

USPC: 220/4.02; 174/650

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. application Ser. No. 17/109,071, filed on Dec. 1, 2020. The content of the application is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to an airtight device and a feedthrough module, and more particularly, to an airtight device of preventing gaseous matter from leakage and a related feedthrough module.

2. Description of the Prior Art

(2) A conventional airtight apparatus puts an object inside an accommodating space of the housing, and a cover is assembled with the housing to seal the accommodating space, so that a liquid cooling device in the airtight apparatus can cool the object; however, liquid matter of the liquid cooling device is transformed into gaseous matter by heat, and therefore the airtight apparatus applies stable pressure for the cover to assemble the cover with the housing, so as to prevent the gaseous matter from leakage. A transmission module may be arranged through the housing via holes or openings formed on the housing, thereby allowing signal/power transmission. However, the opening or the holes may cause liquid/gas leakage from the housing; besides, the holes on the housing may be designed for a specific type of the transmission module, which results in inconvenience when replacing the module of the specific type with a module of another type.

SUMMARY OF THE INVENTION

(3) The present invention provides an airtight device of preventing gaseous matter from leakage and a related feedthrough module for solving above drawbacks.

(4) According to one embodiment of the present disclosure, an airtight device includes a tank and a

feedthrough module. The tank includes a tank body that defines an accommodating space, wherein the tank body includes a lateral wall, wherein an opening is formed on the lateral wall. The feedthrough module is disposed on the opening. The feedthrough module includes a base, a sealing component, a covering component, and at least one transmission component. The base is disposed on the lateral wall, a groove is formed on a surface of the base. A part of the sealing component is disposed inside the groove. The covering component is assembled with the base and adapted to press the sealing component and cover the opening. The at least one transmission component is assembled with the covering component.

(5) According to one embodiment of the present disclosure, a feedthrough module includes a base, a sealing component, a covering component, and at least one transmission component, and at least one transmission component. The base is formed with a groove and a plurality of engaging holes. The sealing component is partially disposed in the groove. The covering component assembled with the base and adapted to press the sealing component, wherein a plurality of fixing holes are formed on the covering component. The at least one transmission component is assembled with the covering component. The plurality of fixing components are adapted to insert into the plurality of fixing holes and engage with the base for uniformly pressing the sealing component.

(6) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram of an airtight device according to an embodiment of the present disclosure.

(2) FIG. 2 is a diagram of a reciprocating engaging mechanism according to an embodiment of the present disclosure.

(3) FIG. 3 is a diagram of a part of the airtight device according to the embodiment of the present disclosure.

(4) FIG. 4 is an exploded diagram of a feedthrough module according to a first embodiment of the present disclosure.

(5) FIG. 5 is an assembly diagram of the feedthrough module according to the first embodiment of the present disclosure.

(6) FIG. 6 is an exploded diagram of the transmission component in another type according to the first embodiment of the present disclosure.

(7) FIG. 7 is an assembly diagram of the transmission component according to the first embodiment of the present disclosure.

(8) FIG. 8 is a sectional view of a part of a covering component and a fixing component in one operation mode according to the first embodiment of the present disclosure.

(9) FIG. 9 is a sectional view of the part of the covering component and the fixing component in another operation mode according to the first embodiment of the present disclosure.

(10) FIG. 10 is an exploded diagram of the feedthrough module according to a second embodiment of the present disclosure.

(11) FIG. 11 is an assembly diagram of the feedthrough module according to the second embodiment of the present disclosure.

(12) FIG. 12 is a sectional view of the feedthrough module according to the second embodiment of the present disclosure.

(13) FIG. 13 is an exploded diagram of the feedthrough module according to a third embodiment of the present disclosure.

(14) FIG. 14 is an assembly diagram of the feedthrough module according to the third embodiment

of the present disclosure.

DETAILED DESCRIPTION

(15) Please refer to FIG. 1. FIG. 1 is a diagram of an airtight device **10** according to an embodiment of the present disclosure. The airtight device **10** can include a tank **12**. The tank **12** comprises a tank body **121**, a tank lid **122**, a liftable hinge mechanism **16** and a reciprocating engaging mechanism **18**. An accommodating space **123** is defined inside the tank body **121**. The tank lid **122** is used to cover the tank body **121**. The liftable hinge mechanism **16** and the reciprocating engaging mechanism **18** can be respectively disposed between the tank body **121** and the tank lid **122**. The tank lid **122** can be connected to the tank body **121** in a rotatable and shiftable manner via the liftable hinge mechanism **16**, which means the tank lid **122** can be liftable relative to the tank body **121**. With the help of the reciprocating engaging mechanism **18**, the tank lid **122** can prevent gaseous matter inside the tank body **121** from leakage. The reciprocating engaging mechanism **18** can simultaneously press and release all sides of the tank lid **122**. One or more electronic apparatus (which is not shown in FIG. 1) such as IT gear can be disposed inside the tank body **121** for cooling. The tank lid **122** can be rotated and lifted relative to the tank body **121** for exposing or sealing the electronic apparatus disposed inside the accommodating space **123**.

(16) Please refer to FIG. 2. FIG. 2 is a diagram of the reciprocating engaging mechanism **18** according to an embodiment of the present disclosure. The reciprocating engaging mechanism **18** can include a shaft **24**, a rotary plate **26**, a first engaging component **28**, a second engaging component **30** and a linkage module **32**. The shaft **24** can be disposed on the tank lid **122**. The rotary plate **26** can be rotatably connected to the shaft **24**. The first engaging component **28** can be connected to the rotary plate **26** via the linkage module **32**. The second engaging component **30** can be disposed on the tank body **121** or the tank **12**. The linkage module **32** can include several structural elements connected between the rotary plate **26** and the first engaging component **28**. Rotation of the rotary plate **26** can drive the linkage module **32** to move and switch the first engaging component **28** between a locking mode and an unlocking mode, so as to tightly shelter the accommodating space **123** by the tank lid **122** for preventing the gaseous matter inside the accommodating space **123** from leakage.

(17) Please refer to FIG. 3. FIG. 3 is a diagram of a part of the airtight device **10** according to the embodiment of the present disclosure. The tank body **121** is configured to accommodate liquid coolant (which is not marked in the figures), and one or more heat generating device **34** (e.g., the electronic apparatus mentioned as above) disposed inside the tank body **121**. During operation, the heat generating device **34** is being immersed by the coolant. The airtight device **10** can further include a power providing module **36**, a condenser **38** and at least one feedthrough module **40** (e.g., the feedthrough modules **40H** and **40L**). The power providing module **36** can be disposed under the heat generating device **34**, and connected between the heat generating device **34** and the feedthrough module **40L** therefore being connected with external power supply (which is not shown in the figures). The condenser **38** can be disposed above the heat generating device **34** and is not immersed inside the coolant. The tank body **121** can include at least one opening **42** formed on a lateral wall **141** (e.g., a backside wall of the sink) of the tank body **121**. A number of the opening **42** can correspond to a number of the feedthrough modules **40H** and **40L**. The feedthrough modules **40H** and **40L** can be disposed on the opening **42**, and have an airtight and waterproof function to avoid the gaseous matter and the coolant from leakage through the opening **42**.

(18) In the illustrated embodiment, the feedthrough modules **40H** are arranged above the condenser **38**. During operation, the liquid coolant would absorb the heat generated from the heat generating device **34** and then vaporize (i.e., the “vapor” in FIG. 3). The upward-moving vapor would mostly contact and being condensed by the condenser **38** before reaching the feedthrough module **40H**, therefore reducing the amount of escaping vapor through the opening **42** when the feedthrough module **40H** is removed (e.g., during maintenance). Likewise, the illustrated tank lid **122** is also located above the condenser **38**, therefore reducing the amount of escaping vapor when the tank lid

122 is opened (e.g., through operating the reciprocating engaging mechanism as shown in FIG. 2). In the illustrated embodiment, the feedthrough modules **40H** are arranged between the tank lid **122** and the condenser **38** along a vertical direction. In some embodiments, the heat generating devices **34** are arranged between the feedthrough module **40** (e.g., the feedthrough modules **40H**) for transmitting signal or power and the feedthrough module **40** (e.g., the feedthrough module **40L**) for transmitting power along the vertical direction.

(19) Please refer to FIG. 4 and FIG. 5. FIG. 4 is an exploded diagram of the feedthrough module **40** according to a first embodiment of the present disclosure. FIG. 5 is an assembly diagram of the feedthrough module **40** according to the first embodiment of the present disclosure. The feedthrough module **40** can include a base **44**, a sealing component **46**, a covering component **48**, at least one transmission component **50** and a plurality of fixing components **52**. The base **44** can be disposed on the lateral wall (e.g., the lateral wall **141** in FIG. 3) of the tank body **121**, and a groove **54** can be formed on a surface **56** of the base **44** opposite to the tank body **121**. The sealing component **46** can be partially disposed inside the groove **54** and be further pressed by the covering component **48**. The covering component **48** can be attached to the base **44** and covers the opening (e.g., opening **42**) of the tank via the plurality of fixing components **52**. The covering component **48** can have a plurality of fixing holes **58** and at least one installing hole **60**.

(20) A number of the transmission component **50** can correspond to a number of the installing hole **60**. The transmission component **50** can be arranged through the installing hole **60** in an airtight and waterproofing manner, and be electrically connected to the heat generating device **34** or the power providing module **36**, which depend on a type of the feedthrough module **40**. A number of the fixing components **52** can correspond to a number of the fixing holes **58**. One of the plurality of fixing components **52** can be inserted into one of the plurality of fixing holes **58**, and be engaged with one of a plurality of engaging holes **64** formed on the surface **56** of the base **44**. Therefore, when the plurality of fixing components **52** is uniformly inserted into the plurality of fixing holes **58** and engaged with the plurality of engaging holes **64**, the sealing component **46** can be uniformly pressed by the base **44** and the covering component **48** for filling the groove **54**, so as to provide the preferred airtight and waterproof function.

(21) In some embodiments, the base **44** is configured to be attached a surface of the lateral wall (e.g., lateral wall **141**) opposite to the accommodating space **123** defined by the tank body **121**, and the covering component **48** is configured to be attached to the base **44** from outside of the tank body **121**. During maintenance, the covering component **48** could be easily removed from outside of the tank body **121** without opening the lid **122** of the tank **12**, therefore reducing the amount of escaping vapor. During maintenance, a dummy cover could be further attached to the base **44**, easily from outside of the tank **12**, such that the opening (e.g., opening **42**) is still being covered while the covering component **48** is removed, therefore reducing the amount of escaping vapor. Besides, the groove **54** on the base **44** is shown to be annular and the engaging holes **64** are located outside of an area of the base **44** surrounded by the groove **54**. Under such arrangement, even though the pressure of the vapor (e.g., “vapor” in FIG. 3) in the accommodating space **123** is high, the vapor could be blocked by the sealing component **46** before reaching the fixing holes **58** of the covering component **44**, thereby significantly reduce the amount of vapor escaping via the fixing holes **58**. In fact, the illustrated sealing component **46** disposed in the groove **54** is uniformly surrounded by the engaging holes **64** and being uniformly pressed.

(22) As shown in FIG. 4, a window **62** can be further formed on the base **44**. The transmission component **50** which arranged through the installing hole **60** can pass through the window **62** to reach into the accommodating space **123** for electrically connecting with the heat generating device **34** or the power providing module **36**. The window **62** can be surrounded by the groove **54**, and the groove **54** can be surrounded by the plurality of engaging holes **64**, so that the feedthrough module **40** can uniformly press the sealing component **46** by moving the covering component **48** toward the base **44** for shortening a distance between the base **44** and the covering component **48**. In the first

embodiment, the feedthrough module **40** is disposed on the opening **42** located above the electronic device (e.g., the heat generating device **34** in FIG. **3**) and the condenser (e.g., the condenser **38** in FIG. **3**), so that the transmission component **50** of the feedthrough module **40** can be electrically connected to the heat generating device **34**. In addition, the covering component **48** can be made of metal material, and be optionally coated by an isolation layer in accordance with the design demand.

(23) Please refer to FIG. **6** and FIG. **7**. FIG. **6** is an exploded diagram of the transmission component **50** in another type according to the first embodiment of the present disclosure. FIG. **7** is an assembly diagram of the transmission component **50** according to the first embodiment of the present disclosure. The transmission component **50** can include a cable **66**, a nut **68** and a sealing element **70**. The cable **66** can have a thread structure **72** and an annular portion **74**. The annular portion **74** can be distant from or adjacent to the thread structure **72**. One end of the cable **66** can insert into the installing hole **60**, and the other end of the cable **66** can extend into the accommodating space **123** for connecting the heat generating device **34** or the power providing module **36**. The nut **68** can be engaged with the thread structure **72** and press one surface **76** (e.g., opposite to the accommodating space **123**) of the covering component **48**. The cable **66** can be locked with the nut **68** and therefore the annular portion **74** can abut toward another surface **78** (e.g., facing the accommodating space **123**) of the covering component **48** opposite to the nut **68**. A sealing element **70** can be disposed between the annular portion **74** of the cable **66** and the surface **78** of the covering component **48** by pressure of the covering component **48** and the annular portion **74** for the preferred airtight and waterproof function, such as the embodiment shown in FIG. **6**.

(24) In the embodiment shown in FIG. **6**, the sealing element **70** is clipped by the covering component **48** and the annular portion **74**. Since the surface **78** of the covering component **48** provides a sufficient contact area for abutting against the sealing element **70**, the sealing element **70** can be made with a sufficiently large size even though a size of the nut **68** is being reduced. By reducing the size of the nut **68**, a number of the installing holes **60** formed on the covering component **48** can be increased, thereby increasing the number of transmission module.

(25) Please refer to FIG. **8** and FIG. **9**. FIG. **8** and FIG. **9** are sectional views of a part of the covering component **48** and the fixing component **52** in different operation modes according to the first embodiment of the present disclosure. The fixing component **52** can include a resilient element **85** (e.g., a spring), a bushing **86** and a screwing body **88**. Two opposite ends of the resilient element **85** can be respectively connected to the bushing **86** and a top of the screwing body **88**. The bushing **86** can be inserted into the fixing hole **58** formed on the covering component **48**, and a protrusion **90** of the bushing **86** can be protruded from the fixing hole **58**. Then, the protrusion **90** can be pressed and deformed by an external force to engage with a depression **92** formed on an inner surface **94** of the fixing hole **58** for a riveting function. In response to engagement of the bushing **86** and the depression **92**, the screwing body **88** can be engaged with the engaging hole **64** of the base **44** and the covering component **48** can be tightly assembled with each other via the fixing component **52** to provide the preferred airtight and waterproof function.

(26) As shown in FIG. **8**, when the resilient element **85** is not compressed, a front end **881** of the screwing body **88** does not protrude from the fixing hole **58** on the covering component **48**, so as to avoid the fixing component **52** and the base (e.g., the base **44** in FIG. **9**) from structural interference. The screwing body **88** cannot be separated from the bushing **86** via connection of the resilient element **85**.

(27) As shown in FIG. **9**, the fixing component **52** is pushed to the left and the resilient element **85** is being compressed. The front end of the screwing body **88** can be engaged with a thread structure **641** of the engaging hole **64** formed on the base **44**. An inner surface of the bushing **86** may not have the thread structure to provide rapid and convenient advantages of engaging the fixing component **52** with the base **44**.

(28) Please refer to FIG. **10** to FIG. **12**. FIG. **10** is an exploded diagram of the feedthrough module

40A according to a second embodiment of the present disclosure. **FIG. 11** is an assembly diagram of the feedthrough module **40A** according to the second embodiment of the present disclosure. **FIG. 12** is a sectional view of the feedthrough module **40A** according to the second embodiment of the present disclosure. In the second embodiment, elements having the same numerals as ones of the first embodiment have the same structures and functions, and a detailed description is omitted herein for simplicity. The feedthrough module **40A** can include the base **44**, the sealing component **46**, a covering component **48A**, a transmission component **50A** and a plurality of fixing components **52A**. The transmission component **50A** can be inserted into and engaged with the installing hole **60** of the covering component **48A** in the airtight and waterproofing manner.

(29) In the second embodiment, the transmission component **50A** can include an array of several cables **80**, a case **82** and a sealing element **84**. An adapter of the cable **80** can be supported by the case **82** connected with the covering component **48A**. The sealing element **84** can be disposed around the cable **80** and filled within a sunken portion **49** of the covering component **48A** for the preferred airtight and waterproof function. The sunken portion **49** can be located on an area of the covering component **48** where the installing hole **60** and the cable **80** is arranged. The sealing element **84** can be epoxy. The feedthrough module **40A** can be disposed on the opening **42** located above the condenser **38**, and the transmission component **50A** of the feedthrough module **40A** can be electrically connected with the heat generating device **34**.

(30) A thickness of the covering component **48A** can be sufficiently large (e.g., greater than a thickness of the covering component **48** in the first embodiment), so that the fixing hole **58** on the covering component **48A** can include an inner thread portion **96** and a non-thread portion **98** adjacent to each other. A length of the non-thread portion **98** of the fixing hole **58** can be greater than a length of an outer thread portion **100** of the fixing component **52A**. A length of the thread structure **641** of the engaging hole **64** is configured greater than the length of the outer thread portion **100** of the fixing component **52A**. The inner thread portion **96** can be used to block the outer thread portion **100** to prevent the fixing component **52A** from being separated from the fixing hole **58**. The outer thread portion **100** can be stayed inside the fixing hole **58** when the fixing component **52A** is disengaged from the engaging hole **64**. A washer **102** can be pressed by the fixing component **52A** to abut against the covering component **48A** for the preferred airtight and waterproof function.

(31) Please refer to **FIG. 13** and **FIG. 14**. **FIG. 13** is an exploded diagram of the feedthrough module **40B** according to a third embodiment of the present disclosure. **FIG. 14** is an assembly diagram of the feedthrough module **40B** according to the third embodiment of the present disclosure. In the third embodiment, elements having the same numerals as ones of the foresaid embodiment have the same structures and functions, and the detailed description is omitted herein for simplicity. The feedthrough module **40B** can include the base **44**, the sealing component **46**, a covering component **48B**, a transmission component **50B** and a plurality of fixing components **52B**. The feedthrough module **40B** can be disposed on the opening **42** located under the heat generating device **34**, and can be electrically connected to the external power supply.

(32) In the third embodiment, the covering component **48B** and the transmission component **50B** can be made of the same conductive material. For example, the transmission component **50B** can be integrated with the covering component **48B** monolithically. The covering component **48B** can be further coated by the isolation layer. The fixing component **52B** can be inserted into the fixing hole **58** on the covering component **48B** and engaged with the engaging hole **64** on the base **44**. The sealing component **46** can be disposed between the base **44** and the covering component **48B**, and can be pressed to fill with the groove **54** on the base **44** when the fixing component **52B** is tightly engaged with the engaging hole **64**.

(33) In conclusion, the feedthrough module of the airtight device in the present disclosure can utilize the plurality of fixing components to lock the base and the covering component, and the sealing component can be located between the base and the covering component and partly inside

the groove on the base. The plurality of fixing components can be separately engaged with the engaging holes on the base in the symmetrical and staggered manner, so that the sealing component can be uniformly pressed and deformed to avoid the gaseous matter in the sink from leakage through unexpected space between the base and the covering component. The fixing component can be designed in accordance with the thickness of the covering component. The transmission component can be designed as a connector, a cable array or a busbar, which depends on an actual demand. The feedthrough module which provides the airtight function can be disposed on the opening above the heating device and not immersed inside the coolant, and the feedthrough module which provides the waterproofing function can be disposed on the opening under the heating device and immersed inside the coolant.

(34) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. An airtight device, comprising: a tank configured to accommodate liquid coolant and related gaseous matter and comprising a tank body that defines an accommodating space, wherein the tank body comprises a lateral wall, and an opening is formed on the lateral wall; and a feedthrough module disposed on the opening, the feedthrough module comprising: a base disposed on the lateral wall, a groove being formed on a surface of the base; a sealing component, a part of the sealing component being disposed inside the groove; a covering component assembled with the base and adapted to press the sealing component and cover the opening; and at least one transmission component assembled with the covering component; wherein the feedthrough module is set on at least one of position arranged above a condenser inside the accommodating space and position located under a heat generating device inside the tank body.
2. The airtight device of claim 1, wherein at least one installing hole is formed on the covering component, and the at least one transmission component is arranged through the at least one installing hole to assemble with the covering component.
3. The airtight device of claim 1, wherein the at least one transmission component is integrated with the covering component monolithically.
4. The airtight device of claim 1, wherein the base is formed with a window surrounded by the groove, the at least one transmission component reaches into the accommodating space via the window and the opening.
5. The airtight device of claim 4, wherein the base is formed with a plurality of engaging holes; the covering component is formed with a plurality of fixing holes corresponding to the engaging holes; the feedthrough module further comprises a plurality of fixing components adapted to insert into the plurality of fixing holes and the plurality of engaging holes so as to engage with the base for uniformly pressing the sealing component.
6. The airtight device of claim 5, wherein the groove is surrounded by the plurality of engaging holes.
7. The airtight device of claim 5, wherein each of the plurality of fixing components comprises a bushing, a protrusion of the bushing is engaged with a depression formed inside an inner surface of each of the plurality of fixing holes.
8. The airtight device of claim 5, wherein each of the plurality of fixing holes on the covering component comprises an inner thread portion and a non-thread portion adjacent to each other, a length of the non-thread portion is greater than a length of an outer thread portion of each of the plurality of fixing components.
9. The airtight device of claim 8, wherein a length of a thread structure of the engaging hole of the

base is configured greater than the length of a thread portion of the fixing components.

10. The airtight device of claim 8, wherein the each of the plurality of fixing components comprises a washer adapted to abut against the covering component.

11. The airtight device of claim 1, wherein the covering component is made of conductive material and further coated by an isolation layer.

12. The airtight device of claim 1, wherein the at least one transmission component comprises a cable and a nut, the cable comprises a thread structure, the cable inserts into at least one installing hole formed on the covering component and extends into the accommodating space, the nut is engaged with the thread structure and adapted to press a surface of the covering component opposite to the accommodating space.

13. The airtight device of claim 12, wherein the at least one transmission component further comprises a sealing element disposed between an annular portion of the cable and a surface of the covering component opposite to the accommodating space.

14. The airtight device of claim 1, wherein the at least one transmission component comprises a cable and a sealing element, the cable is inserted into at least one installing hole formed on the covering component; the sealing element is disposed around the cable.

15. The airtight device of claim 1, wherein the tank further comprises a tank lid, the airtight device further comprises a reciprocating engaging mechanism disposed on the tank lid and adapted to constrain a movement of the tank lid relative to the tank body, the reciprocating engaging mechanism comprises: a shaft disposed on the tank lid; a rotary plate rotatably disposed on the shaft; a first engaging component; a second engaging component disposed on the tank body; and a linkage module connected between the rotary plate and the first engaging component, rotation of the rotary plate driving the linkage module to move and switch the first engaging component between a locking mode and an unlocking mode, so as to tightly shelter the accommodating space of the tank body by the tank lid for preventing the gaseous matter inside the sink from leakage.

16. The airtight device of claim 1, wherein the feedthrough module is arranged above the condenser inside the tank.

17. The airtight device of claim 16, wherein the feedthrough module is arranged between a tank lid of the tank and the condenser inside the tank.

18. The airtight device of claim 1, wherein the base of the feedthrough module is disposed on a surface of the lateral wall opposite to the accommodating space; the covering component is configured to be assembled with the base from outside of the tank body.

19. A feedthrough module, comprising: a base, formed with a groove and a plurality of engaging holes; a sealing component, partially disposed in the groove; a covering component assembled with the base and adapted to press the sealing component, wherein a plurality of fixing holes are formed on the covering component; at least one transmission component assembled with the covering component; and a plurality of fixing components adapted to insert into the plurality of fixing holes and the plurality of engaging holes; and the plurality of fixing components adapted to insert into the plurality of fixing holes and engage with the base for uniformly pressing the sealing component, a length of a thread structure of each engaging hole of the base being configured greater than a length of a thread portion of each fixing component.

20. The feedthrough module of claim 19, wherein the groove is annular; the engaging holes are located outside of an area on a surface of the base surrounded by the groove.
